

agent_sep23

A semantic behavior-cloning policy with a verified deterministic day compiler

Release part	Selected identity
Policy	Submission 56424800 transfer fine-tune; best checkpoint epoch 21
Native model	BCW23004 FP32; 509,359 parameters; 2,037,777 bytes
Compiler	Product-specific collection fallback; source hash ed154431...
Runtime	Collection refinement on; recovery 1; trusted repair; 256 attempts

The learned policy decides what to accomplish today. The compiler decides exactly how to do it legally. This division is the central design choice.

System at a glance

1 OBSERVE	2 PREDICT	3 COMPILE	4 EXECUTE
Public state + own private inventory + causal own-action history	Greedy native FP32 network emits one semantic DayIntent	Bind, fund, allocate, route, and verify a complete day	Run hourly; monitor public changes; repair or act safely

Why this release

This pairing has the completed chain of evidence: immutable model and compiler identities, strict causal data, native parity, compiler tests, fresh full games, local-LB games, and latency. Later variants were not promoted without the same evidence.

The semantic contract

DayIntent contains no routes, tile choices, purchases, hours, or sale quantities. Groups are regenerated from each dawn observation, and members within a group are interchangeable.

Intent family	Exact decision	Compiler responsibility
New crops	Five counts: wheat, carrot, melon, tomato, strawberry	Choose land cells; obtain inputs; schedule planting
Animals	Next-dawn goose/cow/sheep targets	Buy/retain/escape animals; fit structures and capacity
Land	Buy next quadrant today: yes/no	Fund and place the purchase legally
Crop groups	Partition members across reachable biological procedures	Assign cells; schedule water/fertilizer/harvest/retire
Animal groups	Serve count and allowed escape count	Assign animals; schedule feed/care/collection

Binder invariants

- Only dawn intents are accepted; unused slots must be zero.
- Every group partition must exactly sum to its observed size.
- Crop procedures must be biologically reachable; escape and population targets must be feasible.
- Land may be requested only when another quadrant exists.
- Decoder masks enforce counts, terminal-day constraints, and available-space limits before compilation.

Authoritative source: [day_compiler/source/intent.hpp](#)

Contract SHA-256: 544d8ea871fa6ab44794f0c217395f7666e3adf6dfb1f8b255a6575d72940f67

Why semantics, not replay commands

Many legal schedules can realize the same strategy. Training on exact worker commands would punish harmless route differences and force the network to learn legality. Consequence labels preserve strategic supervision while the compiler supplies exact rules, physical assignment, and verification.

Day compiler design

Stage	What it does	Reason
1. Bind + audit	Maps group counts to concrete biological commitments	Reject malformed or impossible intent early
2. Workload	Derives crop/animal work, collection, returns, inputs, capacity, land, hires	Expose the complete mechanical cost
3. Funding	Forecasts cash and market timing; selects sales, buys, and order positions	Prevent nominally good but unfunded schedules
4. Worker search	Assigns tasks and routes under 256 attempts	Bound latency and stay deterministic
5. Verify	Replays the proposed day in the exact engine	Accept only legal, funded, complete schedules
6. Refine	Improves sales, wheat, and product-specific collections	Recover value without breaking commitments
7. React	Monitors observations and repairs hourly	Handle opponent and market changes

What shaped it

- Game rules and engine parity defined legality; route imitation was explicitly not the goal.
- Fixed-task dynamic-programming oracles separated solver defects from impossible workloads.
- Failure repros drove collection, financing, capacity, search-headroom, rival-order, and fallback changes.
- Responsive full games found timing problems that static replay continuations missed.

Sep 23 patch lineage

Escaping collection -> fertilizer collection -> partial recovery -> value collections -> collection hires -> ordinary animals -> ongoing crops -> shed value -> partial input sales -> search headroom -> collection deadlines -> funding-order positions -> product-specific collection fallback.

The selected ed154... compiler beat teammate_shoprouter in 52/64 fresh games at +9,881.58 mean margin. It revised all 16 raw compile failures: zero final uncompiled dawns, missing commitments, or emergencies.

Behavior-cloning policy and training

The selected 509,359-parameter network uses width-128, depth-2 MLP encoders and decoders. It pools product, crop-group, and animal-group representations; board tiles are excluded because placement belongs to the compiler.

Component	Selected design
Inputs	Global state, nine products, crop/animal groups, financial exposure, accounting, available space, causal action history
Coordination	Earlier crop-group plans condition later groups; product-plan summaries condition downstream heads
Crop head	Autoregressive procedure counts with exact partition masks
Animal head	Service counts masked by group size and terminal-day behavior
Global head	Decode land, animal targets, then new crops with prefix and space constraints
Deployment	Greedy temperature-zero FP32 inference from BCW23004; no PyTorch dependency

Two-stage recipe

Stage	Population	Schedule	Selection
Broad parent	Rank <=20: 49,860 train / 10,320 validation dawns	82 epochs; batch 512; AdamW 0.001; seed 2301	Best saved state; validation loss 1.1406101
Style transfer	Submission 56424800: 2,010 train / 570 validation dawns	500 epochs; same optimizer; compiled loss; initialize parent	Epoch 21; validation loss 0.8554325

Epoch 500 reached only 0.2184 training loss but degraded to 2.0465 validation loss. Export best.pt, never the last checkpoint.

Data controls

The frozen corpus has 66,720 dawns from 1,864 episodes and 2,224 perspectives. Whole episodes and both seats stay together under a stable hash split. Future replay actions define labels only. The confirmation reserve remained unopened. Dataset hash: 0ef20bdc...

Quality, performance, and latency

300 / 300	29 / 48	3.791 ms
Native intents matched	Local-LB wins	Mean action latency

Opponent (16 games each)	Wins	Mean margin	Missing
replay-champion	7	+2,740.25	0
ttyn guarded-hysteresis	12	+9,957.63	0
pavel KNN animal rules	10	+3,847.38	0
Overall	29 / 48	+5,515.08	0

Release gates

- 29 native tests passed on the selected compiler lineage.
- C++ matched PyTorch on all 300 parity cases; maximum joint log-probability error was 1.33e-5.
- Strict FP32 was retained: TF32 changed 117 and BF16 changed 1,114 of 10,320 validation intents.
- The package smoke test plays complete 719-action games from both seats with valid actions, deterministic reset, and bounded attempts.

Latency evidence

Across 48 games and 34,512 calls, policy time per game was 1.763-4.325 s (median 2.690 s). Complete matches had median 3.699 s and maximum 5.617 s. The largest action was 762.081 ms; no observed action exceeded one second. Native dawn inference alone measured 1.371 ms over 900 dawns.

Interpretation

Local-LB opponents and seeds were development evidence, not a hidden leaderboard. The fresh compiler confidence interval (+6,231.86 to +13,856.07) supports that compiler change; it does not establish universal opponent performance.

Failure map and next work

Selected replay continuations executed their commitments, yet still trailed the expert. Mean terminal deficit across 72 cases was -22,933.6 from day 12, -12,932.2 from day 20, -5,342.8 from day 25, -2,818.9 from day 27, -1,164.1 from day 28, and -127.5 from day 29. The remaining gap is mostly intent quality and economics.

Failure	Observed evidence	Best next test
Collection + animal service	Delayed feed/care/milk/clipping; one trajectory fed 97 vs 121 and milked 168 vs 203	Capacity-aware intent features; fresh responsive games
Crop decay	Competing work caused large ongoing-crop losses; fixes reduced but did not eliminate them	Explicit decay/capacity pressure and loss-stratified games
Overspending	Episode 112062878 spent 9,172 vs 939 on inputs and finished 32,824 behind the expert	Cash-return features and intent-level budget targets
Funding timing	Rival order positions and sales can break a nominally balanced plan	Condition forecasts on order position; adversarial market games
Greedy crop drift	15,398/18,446 correct with teacher prefixes, 14,537 greedy; 834 later labels unsupported	Train against greedy prefixes or sequence-level feasibility
Fine-tune overfit	Epoch 21 best; epoch 500 validation loss 2.0465	Earlier stopping and larger style cohorts

Promotion rule

Change one layer at a time. Require immutable identities, native parity, both-seat full games, fresh opponents, commitment denominators, and latency before promotion. Keep confirmation episodes and some opponent seeds untouched until the candidate is frozen.

Known evidence gaps

- No protected confirmation estimate has been opened.
- The local-LB set is small and was observed during development.
- Zero missing commitments can still mean a weak intent was executed perfectly.
- The later overflow-return compiler variant lacks enough fresh evidence to replace ed154...

Release references

README.md gives build/use instructions. DESIGN.md is the editable design source. LINEAGE.md records selection history. TRAINING.md is the exact reproduction guide. EVIDENCE.md carries complete metrics and limitations. provenance/ contains the machine-readable identities.